

Homework 06 - The fast Fourier transform

Deadline: 2nd test date (23rd of April)

The mandatory questions account for 100 points, while the optional questions together contribute up to an additional 50 points.

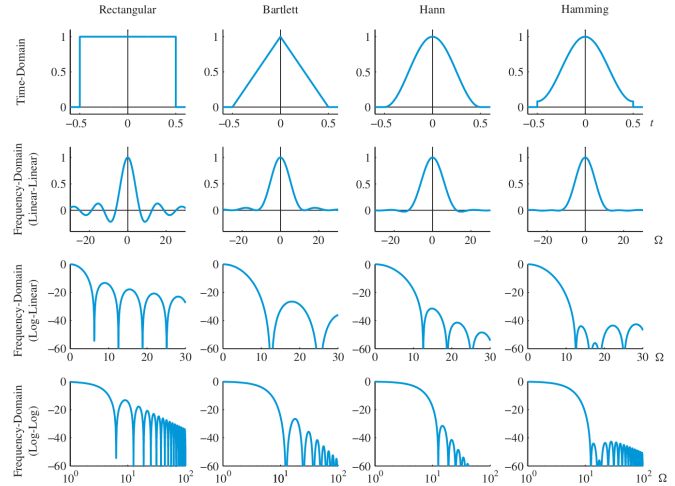
1. Describe the purpose of zero padding in FFT and how it affects frequency resolution.
2. Explain the changes observed in the FFT output when zero padding is applied to a signal.
3. Demonstrate the effect of zero padding.
4. The definition of discrete convolution is

$$z[n] \equiv (x * y)[n] \equiv \sum_{k=-\infty}^{\infty} x[k]y[n-k], \quad n \in \mathbb{Z}.$$

Prove that for the practical case, $x[0], x[1], \dots, x[p-1]$ and $y[0], y[1], \dots, y[q-1]$, this can be rewritten as

$$z[n] = \sum_{k=\max\{0, n-(q-1)\}}^{\min\{p-1, n\}} x[k]y[n-k], \quad n = 0, 1, \dots, p+q-1.$$

- (a) Show that the number of multiplications for calculating the discrete convolution between two sequences of p respectively q elements is $p \cdot q$.
 - (b) Determine the number of additions.
5. Calculate the Fourier transforms of the different windows and reproduce the images below:



6. Demonstrate the effect of leakage and scalloping.
7. Confirm numerically the following properties of the different windows:

Window	Mainlobe width	Rolloff rate (dB/octave)	Peak sidelobe level (dB)
Rectangular	$4\pi/T_0$	-6	-13.3
Bartlett	$8\pi/T_0$	-12	-26.5
Hann	$8\pi/T_0$	-18	-31.5
Hamming	$8\pi/T_0$	-6	-42.9